

Ex (2)

Let a_n and b_n be sequences of real numbers

1) Let a_n and b_n be

$\Rightarrow \sum_{n=1}^{\infty} a_n$ converges.

$$\sum_{n=1}^{\infty} b_n = S \text{ (finite)} : \text{converges}$$

2) Let a_n and b_n be

$$\sum_{n=1}^{\infty} b_n \neq S \text{ (finite)} : \text{diverges}$$

$\Rightarrow \sum_{n=1}^{\infty} b_n$ diverges.

Let a_n and b_n with $a_n \neq 0$.

$$\text{If } \sum_{n=1}^{\infty} a_n \text{ converges} \Rightarrow \sum_{n=1}^{\infty} a_n = 0.$$

$$\sum_{n=1}^{\infty} a_n \neq 0 \Rightarrow \text{If } \sum_{n=1}^{\infty} a_n \text{ diverges.}$$

Ex.

When $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge.

$$1) \sum_{n=1}^{\infty} C a_n = C \sum_{n=1}^{\infty} a_n$$

$$2) \sum_{n=1}^{\infty} (a_n \pm b_n) = \sum_{n=1}^{\infty} a_n \pm \sum_{n=1}^{\infty} b_n.$$